

AI & Backend Engineering Internship Assignment

This document defines the official internship assignment for backend and AI engineering interns. The objective is to evaluate engineering fundamentals, system design, API development, and applied AI skills through progressive, real-world projects.

Overall Objective

Interns will design and implement production-grade backend and AI systems using FastAPI, Python, MySQL, OCR engines, vector databases, and agentic AI frameworks. The assignment progresses from basic backend development to advanced AI orchestration.

Core Technology Stack

- Python 3.x (Object-Oriented Programming)
- FastAPI
- MySQL with SQLAlchemy ORM
- Docker
- Tesseract OCR
- LangChain and LangGraph
- Vector Database (FAISS or Chroma)

Phase 0 — Engineering Foundations

Interns must demonstrate a strong understanding of Python object-oriented programming principles. This phase ensures readiness for subsequent backend and AI development tasks.

Assignment 0.1: Python OOP Mini-System

Implement a basic Document Management domain model using Python classes such as Document, User, and DocumentStore. The design must follow principles of encapsulation, separation of concerns, and clean method design.

Phase 1 — Backend API Development (FastAPI + MySQL)

In this phase, interns will build a RESTful backend service for managing users and documents. The focus is on API design, database modeling, validation, and error handling.

Required API Endpoints

- POST /users — Create a new user
- POST /documents — Create a document
- GET /users/{user_id}/documents — Fetch user documents

Interns must implement proper request validation, HTTP status codes, and database constraints. Unit tests must be written for all endpoints.

Phase 2 — OCR Document Ingestion

Interns will build a service capable of extracting text from scanned documents using OCR. This service must support image and PDF uploads and be fully containerized.

Required Endpoint

- POST /documents/upload — Upload a file and extract text using OCR

The extracted text must be returned in the API response and stored for downstream processing.

Phase 3 — Vector Search & Semantic Retrieval

This phase introduces semantic search. Interns will index extracted document text into a vector database and expose a semantic search API using embeddings.

- POST /documents/index — Index document text
- POST /search — Perform semantic search over documents

Phase 4 — Agentic AI with LangGraph

Interns will design and implement a multi-step AI agent using LangGraph. The agent must classify user intent, retrieve relevant documents, and generate answers.

- POST /ai/ask — Ask questions over documents

Phase 5 — Optional UI Integration

Interns may optionally build a lightweight UI using Streamlit or React to demonstrate document upload and AI querying capabilities.

Evaluation Criteria

- Code quality and architecture — 25%
- API design and correctness — 20%
- AI logic and correctness — 25%
- Testing and validation — 15%
- Dockerization and setup — 15%

All submissions must include a GitHub repository, setup instructions, tests, and a short demo video.